

Information Manifolds: A Framework for Representing Geometry as Information via Curvature-Based Encoding and Information Flow

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Abstract

We introduce a novel framework, termed Information Manifolds, for representing geometric structures as discrete information states derived from local curvature signatures. Unlike classical topology, which focuses on invariants under continuous deformation, our approach captures the evolution of information induced by such transformations.

Given a manifold equipped with a geometric structure, we define a curvature-based encoding that maps local regions into symbolic values, yielding a global information representation of the space. This enables the interpretation of manifolds as information states within a finite symbolic space.

We further define topological information flow as the trajectory of these encoded states under continuous deformation, equipping the framework with notions of distance, velocity, and entropy in information space.

The framework establishes a bridge between geometry, topology, and information theory, and provides a foundation for future extensions, including probabilistic and quantum representations.

1. Introduction

Topology and geometry study structural properties of spaces, often focusing on invariants under deformation. Classical tools such as homology and homotopy classify spaces based on global features.

In contrast, information theory provides tools to encode and measure structure through symbolic or probabilistic representations [1].

Information geometry bridges these domains by equipping spaces of probability distributions with geometric structures such as the Fisher information metric [2, 3]. These statistical manifolds demonstrate that information can naturally induce geometry.

In this work, we reverse this perspective: instead of deriving geometry from information, we encode geometry itself as information.

We introduce Information Manifolds, where geometric structures are mapped into discrete symbolic representations derived from curvature. Rather than studying invariants, we study the evolution of these representations under deformation.

In this work, we do not seek invariants of geometry, but rather its informational evolution.

The overall structure of the proposed framework is illustrated in Figure 1. Figure 1 provides a conceptual overview of the proposed framework, highlighting the transition from geometric structures to discrete information representations, and the emergence of information dynamics under deformation.

2. Information Encoding

Let (M, g) be a Riemannian manifold and $\mathcal{U} = \{U_i\}_{i=1}^N$ a finite cover. Define:

$$I(U_i) \in \{0, 1, \text{NA}\} \quad (1)$$

$$I(U_i) = \begin{cases} 1 & \text{if } \int_{U_i} R(x) dx > \epsilon \\ 0 & \text{if } \int_{U_i} R(x) dx < -\epsilon \\ \text{NA} & \text{otherwise} \end{cases} \quad (2)$$

The information state is:

$$x = (I(U_1), \dots, I(U_N)) \quad (3)$$

3. Information Space

Let:

$$\mathcal{X} = \{0, 1, \text{NA}\}^N \quad (4)$$

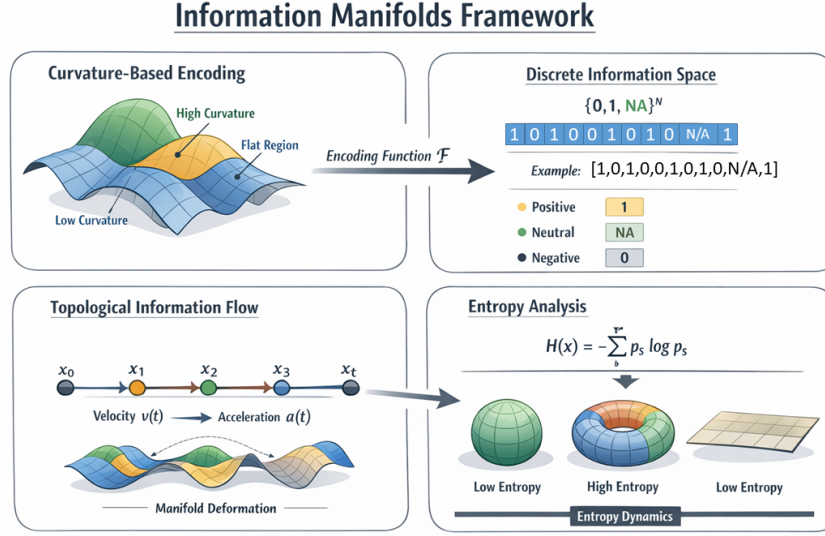


Figure 1: Overview of the Information Manifolds framework. Geometric structures are encoded into discrete information states via curvature-based encoding. These states form elements in a finite symbolic space, where continuous deformation of the underlying manifold induces a trajectory known as topological information flow. The framework further introduces entropy as a measure of structural complexity.

Define Hamming distance:

$$d(x, y) = \sum_{i=1}^N \delta(x_i, y_i) \quad (5)$$

4. Topological Information Flow

Let $\{M_t\}$ be a deformation.

$$x_t = \mathcal{F}(M_t) \quad (6)$$

$$\gamma = (x_0, x_1, \dots, x_T) \quad (7)$$

Velocity:

$$v(t) = d(x_t, x_{t+1}) \quad (8)$$

5. Entropy

$$p_s = \frac{1}{N} |\{i : x_i = s\}| \quad (9)$$

$$H(x) = - \sum p_s \log p_s \quad (10)$$

6. Theoretical Properties

Theorem 1 (Stability). Small curvature perturbations induce bounded changes in encoding.

Theorem 2 (Bounded Distance).

$$0 \leq d(x, y) \leq N \quad (11)$$

7. Discussion

This framework shifts the perspective from invariants to information dynamics. Curvature acts as a local signal that encodes geometric structure into symbolic form.

Unlike classical topology, this representation is intentionally non-invariant, enabling the study of deformation as an information process.

The framework naturally extends toward probabilistic and quantum representations.

8. Numerical Illustration

To demonstrate the proposed framework, we consider a simple discrete approximation of a geometric surface.

Let a 3×3 grid represent local regions U_i , with synthetic curvature values:

$$R = \begin{bmatrix} 0.8 & 0.2 & -0.5 \\ 0.1 & 0 & -0.3 \\ 0.6 & -0.2 & -0.7 \end{bmatrix}$$

Using a threshold $\epsilon = 0.15$, we define the encoding:

$$I(U_i) = \begin{cases} 1 & R > \epsilon \\ 0 & R < -\epsilon \\ \text{NA} & \text{otherwise} \end{cases}$$

This yields the information state:

$$x = \begin{bmatrix} 1 & 1 & 0 \\ \text{NA} & \text{NA} & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Flattening the state:

$$x = (1, 1, 0, \text{NA}, \text{NA}, 0, 1, 0, 0)$$

The empirical distribution over all symbols is:

$$p_1 = \frac{3}{9}, \quad p_0 = \frac{4}{9}, \quad p_{\text{NA}} = \frac{2}{9}$$

This allows defining a full entropy over the entire state.

However, in practical implementations, neutral regions (NA) may be excluded in order to focus on informative curvature signals.

Let:

$$\tilde{x} = \{x_i \mid x_i \neq \text{NA}\}$$

Then:

$$p_1 = \frac{3}{7}, \quad p_0 = \frac{4}{7}$$

The informative entropy is defined as:

$$H_{\text{info}}(x) = - \sum_{s \in \{0,1\}} p_s \log p_s$$

This example illustrates how local curvature variations induce a structured information state, and how different entropy definitions capture distinct aspects of geometric information.

The ratio $\frac{|\tilde{x}|}{N}$ represents the proportion of geometrically active regions.

8.1. Informative Entropy

In practical implementations, neutral regions (NA) may be excluded from entropy calculations, focusing only on informative curvature signals.

Let:

$$\tilde{x} = \{x_i \mid x_i \neq \text{NA}\}$$

Define:

$$p_s = \frac{\#\{i : x_i = s\}}{|\tilde{x}|}, \quad s \in \{0, 1\}$$

Then the informative entropy is:

$$H_{\text{info}}(x) = - \sum_{s \in \{0,1\}} p_s \log p_s$$

9. Conclusion

We introduced Information Manifolds as a framework for representing geometry as information.

This approach provides a new paradigm where spatial structures are interpreted as evolving information systems.

Future work includes probabilistic and quantum extensions.

Appendix A. Computational Illustration

To support reproducibility, we provide a simple numerical implementation of the curvature-based encoding and entropy computation.

In this implementation, neutral regions (NA) are excluded from entropy calculations, focusing only on informative curvature signals.

```
import numpy as np

epsilon = 0.15

R = np.array([
    [0.8, 0.2, -0.5],
    [0.1, 0.0, -0.3],
    [0.6, -0.2, -0.7]
])

def encode(val):
    if val > epsilon:
        return 1
    elif val < -epsilon:
        return 0
    else:
        return "NA"
```

```

vectorized = np.vectorize(encode, otypes=[object])
X = vectorized(R)

filtered_flat = X[X != 'NA'].flatten()

if len(filtered_flat) == 0:
    print("No valid values")
else:
    unique, counts = np.unique(filtered_flat, return_counts=True)
    probs = counts / len(filtered_flat)
    entropy = -np.sum(probs * np.log(probs))

print("Encoded:", X)
print("Entropy:", entropy)

```

This implementation computes the informative entropy, focusing only on regions with non-neutral curvature values.

References

- [1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley, 2006.
- [2] S. Amari, *Information Geometry and Its Applications*, Springer, 2016.
- [3] C. R. Rao, Information and the accuracy attainable in the estimation of statistical parameters, *Bull. Calcutta Math. Soc.*, 1945.
- [4] F. Nielsen, An elementary introduction to information geometry, *Entropy*, 2020.